

SONiX 32-Bit Cortex-M0 Micro-Controller

Flash Memory Controller Example Code

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AMENDMENT HISTORY

Version	Date	Description
1.0	2024/11/29	1. First version.

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1 OVERVIEW

This document provides our customers the C program examples of Flash memory controller.

2 FEATURE

- Page Erase
- Word Program
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.

4 EXAMPLE CODE

4.1 Page Erase

1. Set ISP_TARGET_ADDR as the desired erased Flash address in main.c.

*** Note:**

1. The desired address shall be word-alignment and less than ROM size of used MCU.
2. User shall confirm that there is no any firmware code in the desired Flash address.

For example, to erase Flash address 0x1000, set ISP_TARGET_ADDR as 0x1000.

```
37 //Assign ISP Address
38 #define ISP_TARGET_ADDR 0x1000 //Word-alignment
```

2. Call FLASH_EraseSector function, the page where the address 0x1000 belongs to will be erased, the content of that page will be 0xFF.

```
79 //2. Erase page address @ISP_TARGET_ADDR
80 FLASH_EraseSector(ISP_TARGET_ADDR);
```

*** Note:**

1. Please get the information of the page size from the FLASH chapter in the datasheet of the selected MCU.
2. No need to erase for SN32F700B/760B/770 or SN32F260/240B, these MCU can still be programmed if the ROM is not blank.

4.2 Page Program

1. Please fill in the data which are desired to program to wFLASH_PGRAM[...] in main function, and please declare the proper data size of wFLASH_PGRAM[...].
2. Call FLASH_ProgramPage function, and fill in the proper arguments like Flash address to be programmed (ISP_TARGET_ADDR), total bytes to be programmed, and the address of wFLASH_PGRAM[0], the program process will be processed.
3. The result (OK or FAIL) will be returned.

```
82 //3. Program content of wFLASH_PGRAM[0] & [1] to User ROM address ISP_TARGET_ADDR & (ISP_TARGET_ADDR+4)
83 wFlashBackup[0] = 0xA55A3CC3;
84 wFlashBackup[1] = 0x96691EE1;
85 wFlashBackup[2] = 0xB44BF00F;
86 wFlashBackup[127] = 0x7887D22D;
87
88 if (FLASH_ProgramPage(ISP_TARGET_ADDR, 512, (uint8_t *)wFlashBackup)) == FAIL)
89 {
90     while (1);    //Program Fail
91 }
```

5 RESULT

5.1 Page Erase

The result of FLASH_EraseSector(ISP_TARGET_ADDR) is showed as below in the memory window, the whole page is blank (the contents are all 0xFF).

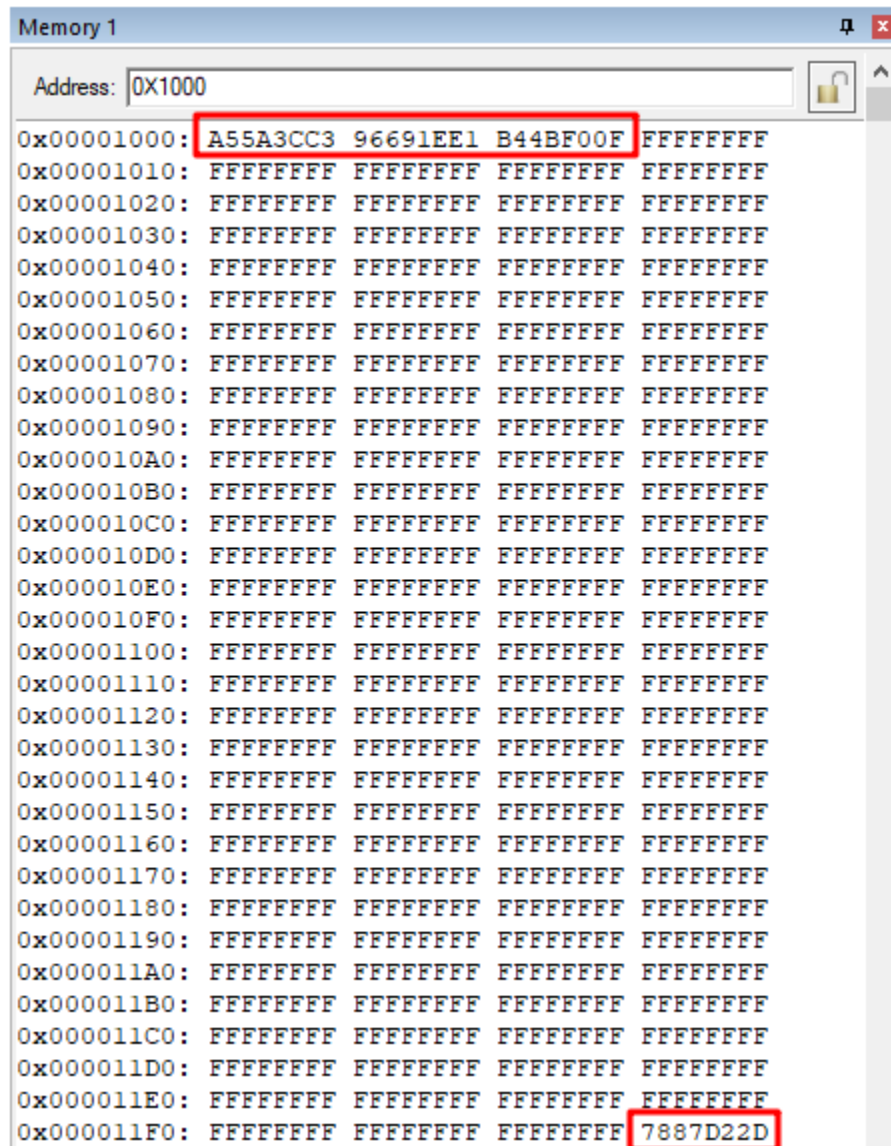
Memory 1	Memory 1
Address: 0X1000	Address: 0X1000
0x00001000: A55A3CC3 96691EE1 B44BF00F FFFFFFFF	0x00001000: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001010: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001010: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001020: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001020: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001030: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001030: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001040: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001040: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001050: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001050: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001060: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001060: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001070: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001070: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001080: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001080: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001090: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001090: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x000010A0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x000010A0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x000010B0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x000010B0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x000010C0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x000010C0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x000010D0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x000010D0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x000010E0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x000010E0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x000010F0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x000010F0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001100: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001100: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001110: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001110: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001120: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001120: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001130: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001130: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001140: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001140: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001150: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001150: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001160: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001160: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001170: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001170: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001180: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001180: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x00001190: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x00001190: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x000011A0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x000011A0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x000011B0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x000011B0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x000011C0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x000011C0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x000011D0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x000011D0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x000011E0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF	0x000011E0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF
0x000011F0: FFFFFFFF FFFFFFFF FFFFFFFF 7887D22D	0x000011F0: FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF

*** Note:**

1. Please get the information of the page size from the FLASH chapter in the datasheet of the selected MCU.
2. Take SN32F700 as example, the page size is 0x200 bytes, since ISP_TARGET_ADDR is 0xF00, in fact, the page (0xE00~0xFFF) which the address 0xF00 belongs to will be erased.
3. Take SN32F100/760/240/780/280/790/290 as example, the page size is 0x400 bytes, since ISP_TARGET_ADDR is 0xF00, in fact, the page (0xC00~0xFFF) which the address 0xF00 belongs to will be erased.

5.2 Page Program

The result of `FLASH_ProgramPage(ISP_TARGET_ADDR, 8, (uint8_t *) (wFLASH_PGRAM))` can be found in the memory window, if the function returns OK, then the content at Flash address 0x1000 to 0x1008 is 0xA55A3CC3, 0x96691EE1, 0xB44BF00F, and the content at Flash address 0x11FC is 0x7887D22D.



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